

Forecast Improvements

We next compared the results from our preferred ARMA model (shown above) to a simple 30-day ADV measure (e.g., ADV30) to determine the extent of the forecasting improvement. The preferred ARMA model reduced forecast error 17% for large cap stocks and 21% for small cap stocks.

Daily Volume Forecasting Model

Our daily volume forecasting models for large and small cap stocks can finally be formulated as:

$$\text{Large Cap } \hat{V}(t) = MDV(10) \cdot \text{Day Of Week}(t) + 0.450 \cdot e(t-1) \quad (7.51)$$

$$\text{Small Cap } \hat{V}(t) = MDV(10) \cdot \text{Day Of Week}(t) + 0.283 \cdot e(t-1) \quad (7.52)$$

Conclusion 3

- There is statistical evidence that there is persistence of volume trading.
- Forecasts can be improved through incorporation of an autoregressive term.
- Table 7.2 shows the one period lag analysis results and the error correlation results. Notice how dramatically the correlation across successive period error terms decreases. The correlation of the error terms decrease -0.311 by using the ARMA model Table 7.3 shows

Table 7.2 One-Period Lag—Error Correlation

Category	Beta	Correlation Before Adj.	Correlation After Adj.	Net Chg. Improvement
LC	0.450	0.452	− 0.015	− 0.437
SC	0.283	0.284	− 0.008	− 0.276
All	0.319	0.320	− 0.010	− 0.311

Table 7.3 Comparison of Error Terms Between ARMA Model and ADV Model

Mkt Cap	ADV30	ARMA	Net Change Improvement	Percent Improvement
LC	40.9%	33.9%	− 7.0%	− 17.1%
SC	56.0%	44.3%	− 11.7%	− 20.9%
All	53.2%	42.8%	− 10.4%	− 19.5%

Forecasting Model: $Y(t) = \text{Median}(10) \cdot \text{DOW} + AR \cdot Y(t-1)$